

SECTION 15847
INDUSTRIAL DAMPERS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Automatic and manual damper devices required for the HVAC systems shown on the drawings and by requirements of this Section
- B. Types of dampers specified in this Section include the following:
 - 1. Manual Volume Dampers.
 - 2. Combination Fire and Smoke Dampers.
 - 3. Curtain Fire Dampers.
 - 4. Ceiling Radiation Dampers.
 - 5. Back Draft Dampers.
 - 6. Gravity Relief Dampers.

1.2 RELATED SECTIONS

- A. Section 15840 - Metal HVAC Ductwork and Accessories.
- B. Section 15840.01_dat - Metal HVAC Ductwork Schedule.
- C. Section 15842 - Process Exhaust Ductwork Schedules. D.
Section 15843 - Metal Industrial Ductwork.
- E. Section 230500 - Common Work Results for HVAC.

1.3 REFERENCES

- A. NFPA 90A - Installation of Air Conditioning and Ventilating Systems.
- B. AMCA 500 - Test Method for Louvers, Dampers and Shutters.
- C. UL 555S - Standard for Safety; Fire Dampers.
- D. SMACNA - HVAC Duct Construction Standard Metal and Flexible – 2005
- E. SMACNA - Round Industrial Duct Construction Standards – 1999
- F. SMACNA - Rectangular Industrial Duct Construction Standards SMACNA - 2004
- G. NFPA 101 - Life Safety Code.

1.4 SUBMITTALS

- A. Submit under provisions of Section 230500 - Common Work Results for HVAC.
- B. Product Data: Provide Manufacturer's product data for equipment required for this project. Submit schedule of dampers showing:
 - 1. UL ratings, leakage, pressure drop, and maximum pressure data.
 - 2. Indicate materials, construction, dimensions, and installation details.
 - 3. Verify performance to NFPA, UL, and applicable codes.
 - 4. Include damper pressure drop data based upon tests and procedures performed in accordance with AMCA 500.

- C. Shop Drawings: Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, location and size of each field connection.
 - 1. Manual volume dampers.
 - 2. Fire damper, smoke damper and combination fire and smoke damper installation including sleeves and duct mounted access doors.
- D. Coordination Drawings: Reflected ceiling plans, drawn to scale and coordinating penetrations and ceiling mounted items. Show ceiling mounting access panels and access doors required to access to duct accessories.

1.5 PROJECT RECORD DOCUMENTS

- A. Record actual locations of all dampers.

1.6 QUALITY ASSURANCE

- A. General: All devices, unless otherwise noted, shall be new and of the best quality.
- B. Provide materials, components, accessories, and fabrication methods for dampers specified in compliance with UL555, NFPA, SMACNA and Building Code Requirements.
- C. Dampers shall be warranted against manufacturing defects.
- D. Dampers shall be tested, rated and labeled in accordance with the latest UL requirements.
- E. Damper pressure drop rating shall be based on tests and procedures performed in accordance with AMCA 500.

PART 2 - PRODUCTS

2.1 EQUIPMENT

- A. HVAC Damper Pressure Classification: meet pressure rating of HVAC duct, as listed in Section 15840 - Metal HVAC Ductwork and Accessories and Section 15840.01_dat - Metal HVAC Ductwork Schedule.
- B. For PROCESS EXHAUST damper pressure classifications, see Section 15842 - Process Exhaust Ductwork Schedules and Section 15843 - Metal Industrial Ductwork. Fume & Ammonia duct system dampers, and blast gate dampers (when shown on drawings) shall conform to their respective individual duct specification and this section. All dampers and blast gates shall meet or exceed the adjoining duct's performance and chemical corrosion properties.
- C. Manual Volume Dampers:
 - 1. Opposed Blade Dampers
 - a. Configuration: Manually operated dampers shall be of the multi-blade, opposed blade type. With the exception of round butterfly type dampers, where indicated, single blade dampers are not acceptable. Where required, duct-mounted dampers shall be fabricated with blades parallel to the shortest side of the ductwork in which they are installed.
 - b. Frames: Frames shall be fabricated of 5" wide by 16 USS gauge (minimum) roll-formed galvanized steel or 2" x 1/2" (min.) galvanized structural channel with integral reinforcing ribs and mounting flanges for ease of installation and sized no greater than 6" in width or 48" in length.

- c. Blade Shafts and Bearings: Blade shafts shall be 1/2" round, hex or square, galvanized or cadmium plated steel and shall operate in nylon or oil impregnated bronze bearings mounted in the damper frame. Round blade shafts are to be through-bolted or through-pinned to the damper blades.
 - d. Linkage: Linkage shall be a galvanized or cadmium plated steel assembly of interconnecting crank arms attached to the shaft of each damper blade. This linkage is to be mounted external to the air stream.
 - e. Damper shaft shall be marked in line with blades so that blade position can be observed.
 - f. Accessories: Each damper shall be furnished with an extra shaft extension, hat channel bracket, manual lever with position indicator and a locking quadrant.
 - g. Manufacturer: Dampers shall be Ruskin Model CD35; or an approved equal.
 - h. Dampers shall have a design maximum operating pressure equal to the duct class for duct in which the damper is installed and shall have a maximum velocity rating that is greater than or equal to maximum air velocity at the point at which the damper is installed in the system.
2. Butterfly Dampers: Where shown on drawings, install butterfly dampers with steel axle, bearings, and locking quadrant with position indicator. Provide low leakage wiper and seals.
3. Organic/VOC & Heat/Gen Duct Dampers
- a. Round Isolation or Control Dampers – Furnish and install, at locations shown on plans, P&IDs, and Damper Schedule, heavy duty industrial grade isolation dampers meeting the following specifications. Dampers shall be butterfly type consisting of circular blade, mounted to axle within formed flanged frame. Frame shall be constructed of steel channel with a clean and smooth interior surface. Dampers with full circumference blade stops are not acceptable. Blade shall be minimum 1/4" thick and be complete with a full circumference elastomer type wiper seal retained by a seal ring bolted to the blade. Adhesive type seals are not acceptable. The seal shall be field adjustable and replaceable. Damper shaft shall be continuous, solid cold rolled steel extending through the entire damper diameter and extending beyond damper bearing a minimum of 6". Stub type axles are not acceptable. The axle shall be supported in sealed, relubricable ball bearings mounted outboard of frame and be complete with axle shaft seals. Damper frame and blade shall be fabricated from hot rolled steel. Damper shall also be compatible with continuous exposure with VOC fumes, as applicable, per Chemical Fume Concentrations schedule in ductwork specification and concentrated liquid flowing along duct bottom. For dampers located inside the cleanroom, parts not otherwise protected shall be given one prime coat and one finish coat of bonded industrial alkyd-epoxy enamel. Leakage through damper in closed position shall not exceed .029 SCFM per inch of blade circumference at a pressure differential of 10" W.G. Maximum pressure drop across a 56" diameter unit shall not exceed .015" W.G. at 40,000 CFM. Damper design shall allow application in system with duct static pressures and damper classification per 15842 - Process Exhaust Ductwork Schedule. Submittal shall include published performance data on a complete range of damper sizes developed from testing in accordance with AMCA Standard 500 in an AMCA registered laboratory. Install dampers round and free from distortion with axle running horizontally. If dampers are automatically controlled with damper motor, damper contractor to provide "jack shaft" extension with

spline or key as required for complete assembly. Damper shall be Ruskin model CDRI92, CDR92, or Engineer Approved Equal.

- b. Rectangular Isolation or Control Dampers - Furnish and install, at locations shown on plans, P&IDs, and per Damper Schedule, heavy duty opposed blasé airfoil blade industrial grade isolation/balancing/control dampers meeting the following specifications: Damper frame shall be minimum 8" deep x 2" x 12 gage galvanized steel channel. Blades shall be double skin airfoil design, minimum 14 gage galvanized steel. Axle material shall be 304 SS rod. Bearings shall be stainless steel sleeve pressed into frame. Oil impregnated bronze, synthetic, or bolt on style are not acceptable. Linkage shall be located in jamb out of airstream and constructed of minimum 10 gage steel double clevis arms pivoting on minimum 3/8" diameter stainless steel pivot with lock type retainers. Face linkage in airstream is not acceptable. Standard construction shall include locking hand quadrant for manual operation or crank lever for motor operation as required on the drawings. Dampers shall be equipped with blade and jamb seals for low leakage application and be complete with appropriate blade seals for air stream contaminants. Blade seals shall be mechanically attached to blade. Adhesive type seals are not acceptable. Jamb seals shall be flexible stainless steel located between blade edge and jamb for maximum sealing compression. Wind stops or sponge seals are not acceptable. Damper design shall allow application in system with duct static pressures and damper classification per 15842 - Process Exhaust Ductwork Schedule. Submittal data must include published leakage, pressure drop, and maximum pressure data based on AMCA Standard 500 testing. Data shall be for a full range of damper sizes. Data from one size sample is not acceptable. Leakage shall not exceed 4 CFM per square foot at 1" SP. Bolt holes spacing shall be per PS 15-69. Damper finish shall be bonded industrial polyurethane epoxy enamel in cleanroom areas and galvanized steel everywhere else. Damper shall also be compatible with continuous exposure with VOC fumes, as applicable, per Chemical Fume Concentrations schedule in ductwork specification and concentrated liquid flowing along duct bottom. Contractor shall submit manufacturer's product data, including leakage, pressure drop, and maximum pressure data in accordance with AMCA 500 testing. Submit data for full range of damper sizes. Maximum air pressure drop across 100% open damper shall not exceed 0.15" SPWC at 3,000 fpm velocity. Maximum leakage rate based on blade circumference: 4 cubic feet per minute/sf @ 15" SPWC pressure differential. If dampers are automatically controlled with damper motor, damper contractor to provide "jack shaft" extension with spline or key as required for complete assembly. Damper shall be Ruskin model CD80AF3, CD80AF4, or Engineer Approved Equal.

D. Combination Fire and Smoke Dampers:

1. Multi-blade, horizontal shaft type designed for automatic operation. Lever shall be parallel to airflow when damper is in the open position.
2. Configuration: Dampers shall be provided with opposed blade operation.
3. Frames: Frames shall be 5 inch x 16 gauge roll formed galvanized steel hat-shaped channel reinforced at corners..
4. Blades: Blades shall be air-foil-shaped, single piece, double skin, opposed, minimum 14 ga. equivalent thickness, maximum 6".
5. Axle and Bearings: Axle shall be 1/2" square, hex or otherwise keyed to the damper blade in such a manner as to ensure positive locking connections. Bearings shall be self-lubricating stainless steel sleeve, turning in extruded hole in frame.

6. Linkage: Linkage shall be concealed in frame.
7. Seals: Dampers shall be equipped with extruded vinyl or inflatable silicone fiberglass material blade edge and flexible metal compression jamb.
8. Allowable Leakage: Leakage shall be less than 4.0 CFM/sq. ft. when a 48" x 48" damper section is tested in accordance with AMCA STD. 500 at a static pressure differential of 4" W.G. Results of this test shall be certified by an independent testing laboratory.
9. Allowable Pressure Loss: When tested in the full-open position, the pressure loss of a 48" x 48" damper section shall not exceed 0.2 W.G. at a face velocity of 3000 FPM. Results of this test shall be certified by an independent testing laboratory.
10. Dampers shall be UL Labeled in accordance with Standard 555S, "Standard for Leakage Rated Dampers for Use in Smoke Control Systems", of Underwriter's Laboratories, Inc. Combination fire and smoke dampers shall also be UL-labeled for 1-1/2 hour rating according to UL Standard 555 "Standard for Fire Dampers".
11. Manufacturer: Dampers shall be Ruskin Model FSD60 for combination fire and smoke dampers; or an approved equal.
12. Mounting: Vertical or horizontal.
13. Temperature release device: Heat actuated, quick detect, 165° F. rated.

E. Curtain Fire Dampers (Dynamic)

1. 1-1/2 Hour Labeled Fire Dampers
 - a. Curtain Fire Dampers shall be dynamic type capable of normal operation while HVAC systems remain in operation..
 - b. Frame: Curtain type with blades outside the air stream, fabricated of roll formed galvanized steel with mitered and interlocking corners.
 - c. Dampers shall also be UL-labeled for 1-1/2 hour rating according to UL Standard 555S "Standard for Fire Dampers".
 - d. Provide galvanized steel mounting sleeve.
 - e. Mounting: Vertical or horizontal as indicated.
 - f. Horizontal dampers to include blade lock and stainless steel closing spring.
 - g. Fusible link: Replaceable 165 ° F. rated.
 - d. Manufacturer: Dampers shall be Ruskin Model FSD60 for combination fire and smoke dampers; or an approved equal.

F. Ceiling Radiation Dampers (Dynamic)

- a. 1-1/2 Hour Labeled Fire Damper
 - a. Dampers shall be UL Labeled in accordance with Standard 555S, "Standard for Leakage Rated Dampers for Use in Smoke Control Systems", of Underwriter's Laboratories, Inc.
 - b. Manufacturer: Dampers shall be Greenheck Model CRD-1 with 165 ° F UL listed link or approved equal.

G. Back Draft Dampers: Provide multi blade, parallel action gravity balanced damper with blades of maximum 6 inch width with sealed edges. Damper to be factory assembled in a rattle free manner with 90 degree stop, steel ball bearings and axles. Damper to have an adjusting device to allow opening at a required differential pressure setting.

1. Provide back draft dampers where shown on drawings.
2. Maximum pressure rating for the damper selected shall be at least 150% of the duct pressure class for the duct in which installed.

3. Operating velocity through the damper shall not be greater than 80% of the maximum rated velocity for the damper selected.
- H. Gravity Relief Dampers:
1. Room Relief Dampers
 - a. Provide multi blade, counter balanced back-draft damper, parallel action, damper with blades of maximum 6 inch width with sealed edges.
 - b. Damper to have adjustable counterweights to allow setting the damper to open at an operating static pressure between 0.01"-0.05" w.c.
 - c. Damper to be factory assembled in a rattle free manner with 90 degree stop, steel ball bearings and axles.
 2. Duct Pressure Relief Dampers (Recovery Air System)
 - a. Provide multi blade, counter balanced pressure relief damper, parallel action damper with blades of maximum 6 inch width with sealed edges.
 - b. Damper to have adjustable counterweights to allow setting the damper to open at an operating static pressure between 2.5" - 3.0" w.c.
 - c. Damper to be factory assembled in a rattle free manner with 90 degree stop, steel ball bearings and axles.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Dampers are to be installed where shown on the Drawings or identified in the specifications. Installation is to be neat, workmanlike and in accordance with the instructions of the damper manufacturer.
- B. Fire damper installation shall meet NFPA 90A, UL555 and Building Code requirements.

3.2 COMPLETED WORK

- A. Position each manual operator so that it clearly and accurately indicates the position of the damper blade.
- B. Position access doors in ductwork, sized for maintenance access, adjacent to Fire Dampers and Combination Smoke and Fire Dampers in ductwork where installed.

END OF SECTION